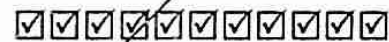


Comprehend to the various design philosophies used for design of reinforced concrete.

Analyze and design the reinforced concrete slab using limit state and working state method.

Analyze and design the reinforced concrete beam using limit state and working state method.

Analyze and design the reinforced concrete column using limit state and working state method.



BTCVC602 Foundation Engineering

Teaching Scheme: (3 Lectures + 1 Tutorial) hours/week

Course Contents

Module 1:

(6 Lectures)

Introduction, General requirements to be satisfied for satisfactory performance of foundations, Soil exploration: Necessity, Planning, Exploration Methods, Soil Sampling Disturbed and undisturbed, Rock Drilling and Sampling, Core Barrels, Core Boxes, Core Recovery, Field Tests for Bearing Capacity evaluation, Test Procedure & Limitations

Module 2:

(7 Lectures)

Bearing Capacity Analysis - Failure Modes, Terzaghi's Analysis, Specialization of Terzaghi's Equations, Skempton Values for No. Meyerhof's Analysis, I.S. Code Method of Bearing Capacity Evaluation, Effect of Water Table, Eccentricity of load, Safe Bearing Capacity and Allowable Bearing Pressure, Settlement Analysis: Immediate Settlement - Consolidation Settlement, Differential Settlement, Tolerable Settlement, Angular distortion

Module 3:

(5 Lectures)

Foundations for Difficult Soils - Guidelines for Weak and Compressible Soils, Expansive soil, Parameters of Expansive Soils, Collapsible Soils and Corrosive Soils, Causes of Moisture changes in Soils, Effects of Swelling on Buildings, Preventative

Measures for Expansive Soils, Modification of Expansive Soils, Design of Foundation on Swelling Soils, Ground Improvement Methods: for general considerations, for Cohesive Soils, for Cohesionless Soils,

Module 4:

(5 Lectures)

Shallow Foundations: Assumptions & Limitations of Rigid Design Analysis, Safe Bearing Pressure, Settlement of Footings, Design of Isolated, Combined, Strap Footing (Rigid analysis), Raft Foundation (Elastic Analysis), I. S. Code of Practice for Design of Raft Foundation

Module 5:

(7 Lectures)

Deep foundations: Pile Foundation: Classification, Pile Driving, Load Carrying Capacity of Piles, Single Pile Capacity, Dynamic Formulae, Static Formulae, Pile Load Tests, Penetration Tests, Negative skin Friction, Under Reamed Piles, Group Action of Piles,

Caissons Foundations: Box, Pneumatic, Open Caissons, Forces, Grip Length, Well Sinking, Practical Difficulties And Remedial Measures

Sheet Piles: Classification, Design of Cantilever Sheet Pile in Cohesionless and Cohesive soils, Design of Anchored Sheet Pile by Free Earth Support Method, Cellular Cofferdams: Types, Cell Fill Stability Considerations

Module 6:

(6 Lectures)

Slope Stability: Different Definitions of Factors of Safety, Types of Slope Failures, Stability of an Infinite Slope of Cohesionless Soils, Stability Analysis of an Infinite Slope of Cohesive Soils, Stability of Finite Slopes- Slip Circle Method, Semi Graphical and Graphical Methods, Friction Circle Method, Stability Number: Concept and its use

Text Books

- Kasamalkar, B.J., "Foundation Engineering", Pittsburgh vintage Grand Prix
- Murthy V.N.S., "Soil Mechanics and Foundation Engineering", CRC Press 2002
- Arora K.R., "Soil Mechanics and Foundation Engineering", Standard publication 2009
- Punmia B. C., "Soil Mechanics And Foundation Engineering", Laxmi publication 16th 2017
- Nayak N.V., "Foundation Design Manual", Dhanpat Rai And Sons
- Brahma S.P., "Foundation Engineering", Tata McGraw-Hill 5th Edition
- Braja Das, "Principles of Geotechnical Engineering", Engage Learning 9th edition
- Bowles J.E., "Foundation analysis & Design", McGraw-Hill Higher Education 5th edition

References Books

- Teng W.C., "Foundation Design", Prentice-Hall Inc
- Tomlinson M.J., "Foundation Design & Construction", Prentice-Hall; 7th edition
- Lee, "Sheet Piles" Concrete Publication, 1961
- Relevant Publications by Bureau of Indian Standards, New Delhi
- IS 6403:1981, IS 1904:1986, IS 4091:1979

Course Outcomes: On completion of the course, the students will be able to:

- To predict soil behavior under the application of loads and come up with appropriate solutions to foundation design queries.
- Analyze the stability of slope by theoretical and graphical methods.
- Analyze the results of in-situ tests and transform measurements and associated uncertainties into relevant design parameters.
- Synthesize the concepts of allowable stress design, appropriate factors of safety, margin of safety, and reliability.



Unit -1

subsurface Investigations For Foundations

contents -

1. Concept of subsurface exploration / Investigation, its purpose and planning.
2. Methods of Investigation.
3. Concept of sampling
4. Field tests
5. Concept of core logging and site investigation report.

Book used - Punmia B.C., "soil Mechanics And Foundation Engineering",
Laxmi publication.

1. Concept of subsurface exploration / Investigation, its purpose and planning.

ue.1 Explain in detail " soil exploration?

2 Explain in detail " soil exploration for an important building project.

3 Explain the steps included in soil exploration.

4 What are the factors affecting cost of soil investigations

→ • Soil exploration / Investigation is nothing but the process of collecting information about the subsurface soil and ground water conditions.

• For bigger sites like dam, bridge, tunnels, highways etc it is necessary that proper soil exploration is carried out.

• Purpose of soil exploration -

i) selection of type and depth of foundation for a given structure.

ii) Determination of the bearing capacity of the soil.

iii) selection of suitable construction technique.

iv) Establishing the ground water table level

v) Determination of suitability of the soil as a construction material.

vi) Such exploration will enable the designer to provide economical design.

- Steps included in soil exploration -

Sub-surface explorations are generally carried out in three steps or stages.

i) Reconnaissance:

It is first step in a sub-surface exploration. It consist of a visit to the site and to study the maps and other necessary records.

ii) Preliminary Explorations:

The basic aim of preliminary exploration programme is to be determine the depth, thickness, extent and composition of each soil layer at the site. Along with above parameters ground water table and depth of rock is also determined. The preliminary explorations are done in the form of trial pits or borings.

iii) Detailed Explorations:

The objective of the detailed explorations is to be determine the engineering properties of the soils. It consist of ~~sampling of~~ testing of the samples in a laboratory.

To determine the properties of the soils in natural state, field tests such as vane shear test, plate load test etc are conducted.

- Factors affecting cost of soil Investigations

i) Non uniformity of layers -

Erratic deposits need more exploration cost than homogenous deposits.

ii) Undisturbed Sampling

Undisturbed sampling needs heavy expenditure.

iii) Depth of explorations

cost increases with increase in depth of exploration.

iv) Nature of project

For earthen dam, the construction cost is low but has high total cost of exploration, while retaining wall have high cost of construction and low exploration cost.

2. Methods of Investigation/Exploration

Que-5 Explain with neat sketch seismic Refraction method of soil Exploration.

6 Explain electrical resistivity method with respect to, Principle, procedure, sketch and limitations.

7 Define boring. Explain in details percussion drilling.

8 Explain electrical resistivity method. Also write application of this method.

9. Describe various methods of drilling holes for subsurface investigation.

10. Explain with neat sketch "wash boring".

11. Explain with neat sketch "Auger boring".

→ The various methods of soil exploration can be as follows;

i) Open excavation

ii) Boring

iii) Geophysical methods;

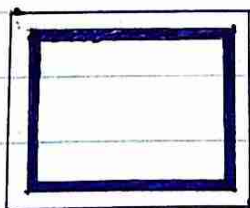
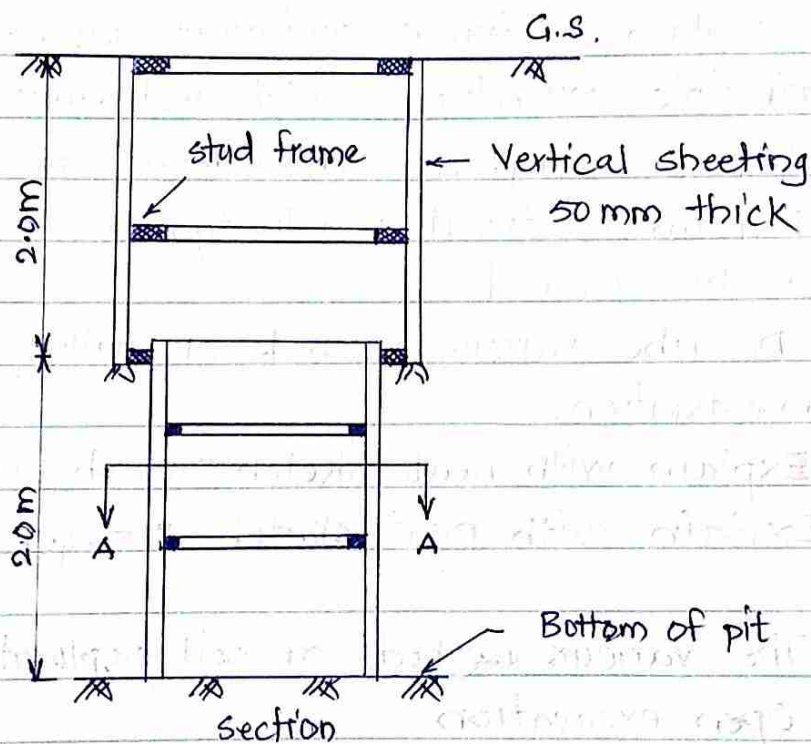
(i) Open excavation -

Open excavation method for sub-surface exploration is adopted to inspect the substrata. Open excavation for soil exploration can be done by two ways;

1. Pits and trenches:

- Pits and trenches are excavated at site for inspection of strata. The pit and trench size should be such as to provide necessary working space.
- According to IS: 4453-1967 a clear working space of $1.2\text{m} \times 1.2\text{m}$ should be provided at the bottom of the pit.
- The shallow pits do not require lateral supports, if depth is up to 3m . If pits are deeper and below ground water table, they require lateral support.
- ~~The~~ Trenches can be defined as 'long shallow pits'. A trench is continuous over a considerable length.

2



Section A-A

2. Drifts and shafts

- Drifts can be defined as horizontal tunnels made in the hill side to determine the nature and structure of the geological strata.
- According to IS 4453-1980 a drift should have the minimum clear dimension 1.5m wide and 2.0m in height in hard rock.
- Large size vertical holes made in the geological formation are called shafts.

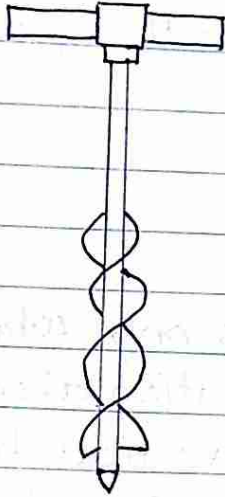
2. Boring -

- Boring are used for explorations only when depth of exploration is large. The information about sub-soil strata is obtained by drilling a vertical bore hole in the ground. Samples are extracted from bore holes and tested in a laboratory.
- Following methods can be used for drilling bore holes depending up on the type of soil and the purpose of boring
 1. Auger boring
 2. Wash boring
 3. Percussion drilling
 4. core drilling.

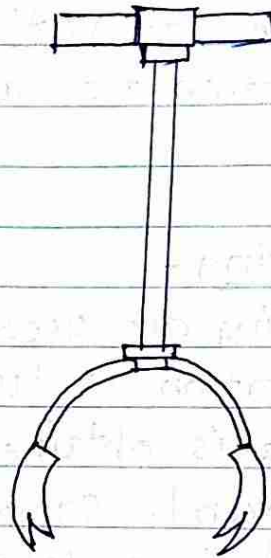
1. 'Auger boring' -

- In cohesive and other soft soils, above water table, augers are used.
- An auger is boring tool, similar to one used by a carpenter for boring holes in a wood.

- Augers can be operated mechanically or manually. Hand augers used for depths up to about 6m and mechanically operated augers are used for greater depth.
- Hand augers are used in boring are about 15 to 20 cm in dia.



Helical auger

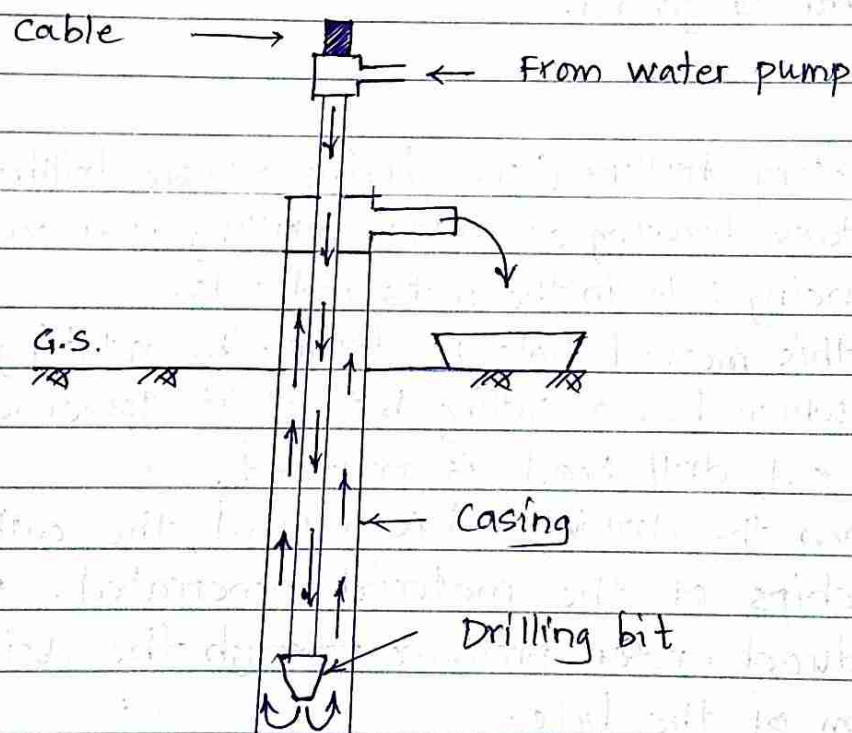


Post-hole auger

- Mechanical augers are driven by power. If depth of bore holes increases above 12m then mechanical auger become inconvenient and other boring methods are used.
- Auger boring becomes trouble-some when there are large boulders, cobbles and other obstructions which prevent drilling of hole.
- The site investigations are done quite rapidly and economically by using auger boring.
- For subsurface investigations of highways, railways and air fields, auger boring is used because depth of exploration is small.

2. Wash boring:

- Wash boring is fast and simple method for drilling holes in all types of soils.
- In wash boring a hole is drilled by driving a casing about 2 to 3 m deep and then inserting into it a hollow drill rod having a chisel-shaped chopping bit at its lower end.
- Water is pumped down the hollow drill rod which is called as 'Wash Pipe'. As a result water comes as a strong jet through a small opening of the chopping bit. By a combination of chopping action and the jetting action the hole is drilled.
- The drilling bit and water jet together disintegrate the soil. The chopped particles are thrown upward through the space b/w the drill rod and the casing.



- As there is complete breakdown of soil particles by chopping action, the wash samples collected in the tub do not represent the soil in its true form.

The samples collected do not have much practical use. But some idea about change in strata can be obtained by the reaction of chopping bit as the hole is drilled. Change in colour of washed water also suggests change in strata.

Advantages -

- i) Instrument used is relatively light in weight
- ii) Inexpensive method of boring
- iii) Fast and simple method.

Disadvantages -

- i) It is slow in stiff and coarse grained soils.
- ii) Cannot be used in rocky strata.
- iii) Good quality undisturbed samples can not be obtained.
- iv) Not suitable in areas where ground water table is very near to ground.

3. Rotary drilling (core boring or core drilling):

- Rotary boring or rotary drilling is a very fast method of advancing hole in the rocks and soils.
- In this method hole is drilled by rotating a hollow drill rod which has a cutting bit at its lower end. At the top of drill rod, drill head is provided.
- When the drilling rod is rotated, the cutting bit shears off chips of the material penetrated. A drilling fluid is introduced under pressure through the drilling rod to the bottom of the hole.
- The cuttings of the material penetrated are carried to the ground by the fluid through the space b/n the drilling rod & the walls of the hole. Another function of fluid is to cool the drilling bit.

Advantages -

1. Rotary drilling can be used in clay, sand and rocks.
2. Bore holes of dia. 50 mm to 200 mm can be easily drilled by this method.

Disadvantages -

If soil is containing a large percentage of particles of gravel size and larger, this method is not well adapted. Because the particles of this size start rotating below the drill rod and it becomes difficult to drill the hole.

4. Percussion Drilling

- This method is specially used for making holes in rocks, boulders and other hard strata.
- Percussion drilling consists of lifting and dropping of a heavy chisel in vertical hole. The material gets pulverised. water is added to the hole, if the chisel strikes above water table. The water forms a slurry of disintegrated material, which is removed by sand pump or a boiler at intervals.
- Casing may be required for this method.

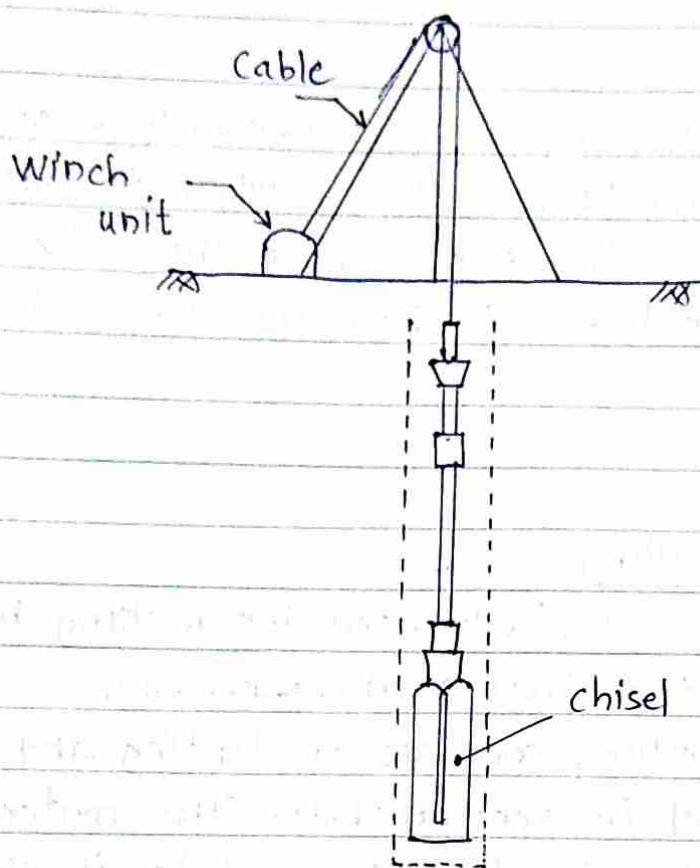
Advantages:

1. Percussion drilling can be used for all types of materials.
2. It is very much useful for boring holes in glacial hills containing boulders.

Disadvantages:

1. Because of heavy blows of the chisel, the material at the bottom is in disturbed state.

2. More expensive as compared to other methods.
3. Difficult to detect minor changes in the properties of the strata penetrated.



5. Core drilling:

- This method is used for advancing holes and for obtaining rock cores. It consists of a core barrel fitted with a drilling bit and is fixed to a hollow drilling rod.
- The bit advances and cuts an annular hole around a core, as the drilling bit is rotated. Then the core is removed from its bottom and is retained by a core lifter and brought to ground surface.
- To keep drilling bit cool and to carry the disintegrated material to the ground surface; water is pumped continuously into the drilling rod.

Geophysical Methods

1. Seismic refraction method

Principle: When a shock or impact is made at a point or in the earth, the resulting seismic (shock or sound) waves travel through the surrounding soil at speeds related to their elastic characteristics.

The velocity is given by,

$$V = C \sqrt{\frac{E \times g}{\gamma}}$$

where;

V = Velocity of the shock wave

E = Modulus of elasticity of soil

g = Acceleration of soil

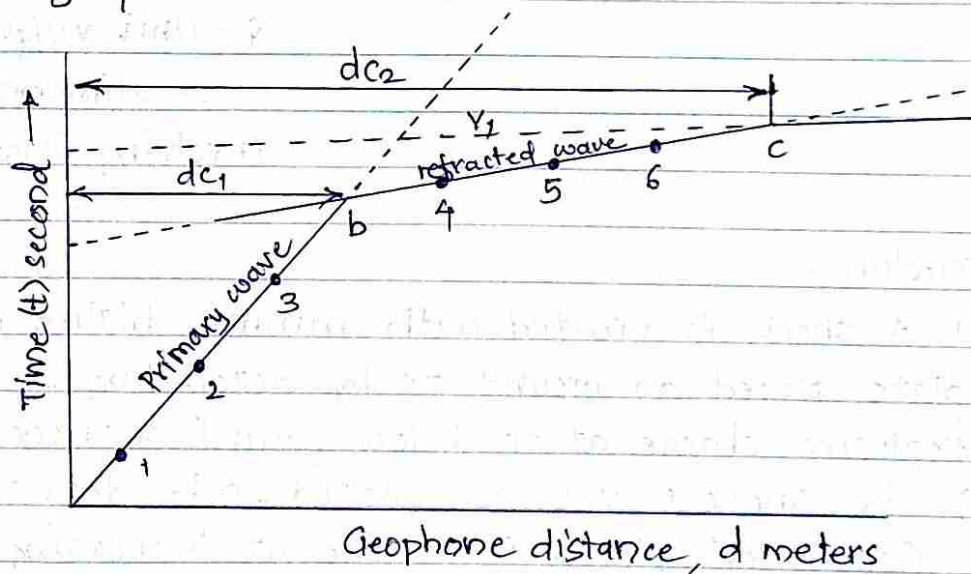
γ = Unit weight of soil.

C = A dimensionless constant involving Poisson's ratio.

Procedure:

- i) A shock is created with hammer hitting a strike plate placed on ground or by detonating a small explosive charge at or below ground surface.
- ii) The shock waves are picked up by detectors called 'Geophones' placed in a line at increasing distances $d_1, d_2, d_3, \dots$ from the origin of the shock.
- iii) The time required for the elastic wave to reach each geophones is automatically recorded by a 'Seismograph'.
- iv) Some of the waves known as Primary waves, travel directly from the source along the ground surface or through the upper strata and are recorded by first geophone.

- v) If sub-soil consist of two or more strata then primary waves travel downwards to lower layer and get refracted at the surface. These waves are known as refracted waves. If underlying layer is denser then these waves travel faster. Refracted waves reach the geophones earlier than the primary or direct waves.
- vi) The distance of the point at which the primary and refracted waves reach the geophone simultaneously is called 'Critical Distance', critical distance depends upon ratio of depth & velocity of strata.
- vii) Results are plotted as distance of travel versus time graph.



- viii) The reciprocal of slope of above travel time graph gives the velocity of the wave.
- ix) Thickness of upper layer is given by,

$$H_1 = \frac{dc_1}{2} \sqrt{\frac{(V_2 - V_1)}{(V_2 + V_1)}} \quad \text{and} \quad H_2 = 0.85 H_1 + \frac{dc_2}{2} \sqrt{\frac{(V_3 - V_2)}{(V_3 + V_2)}}$$

where; dc_1 & dc_2 = critical distances

V_1 = velocity of in upper soft layer of thk (H_1)

V_2 = Velocity in lower hard layer of thk H_2

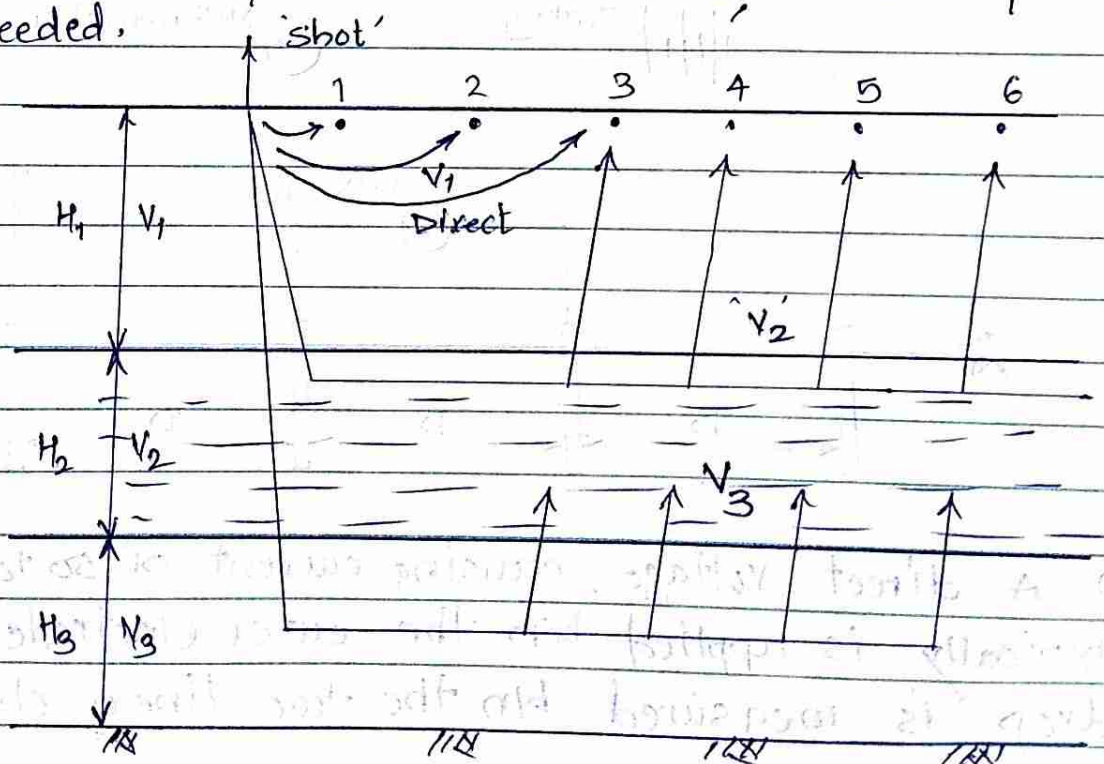
V_3 = Velocity in bottom most layer of thk H_3

H_1 = Thickness of first layer

H_2 = Thickness of second layer.

Limitations -

- i) This method cannot be used if a hard layer overlies a soft layer.
- ii) The method can not be used for the areas covered by concrete asphalt pavements or any other artificial hard material.
- iii) If sub-soil contains buried conduits, irregularly dipping strata and irregular water table, result interpretation becomes very difficult.
- iv) In case of soil having frozen soft surface this method is not useful.
- v) sophisticated and costly equipments are required.
- vi) For interpretation and results, skilled experts are needed.



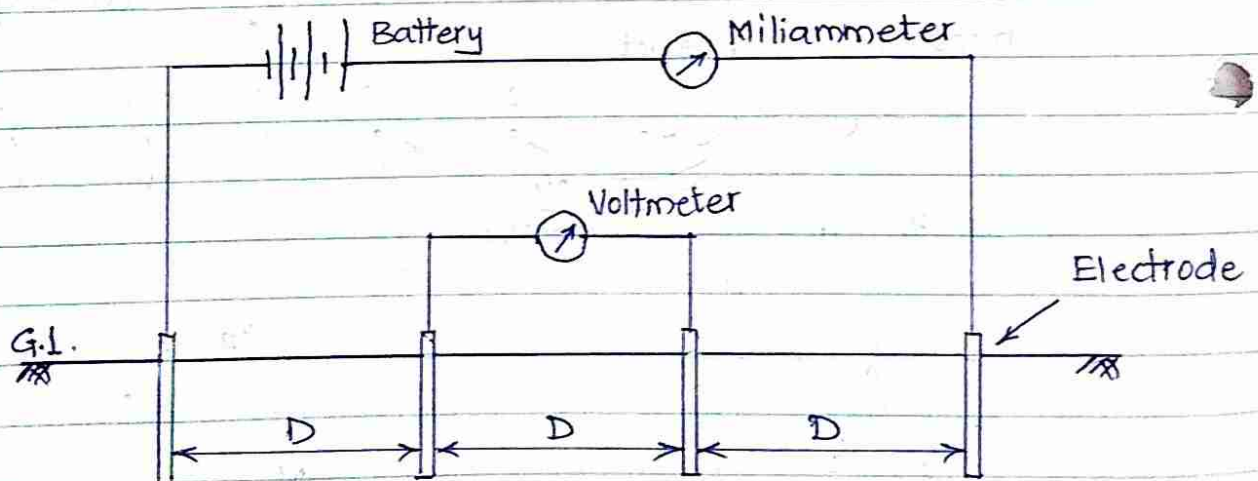
Travel of primary and refracted waves.

2. Electrical Resistivity Method:

Principle: All materials including soil possess resistivity. This method is based on the fact that in soil and rock materials the resistivity values differ sufficiently to allow that property to be used for identification purpose.

Procedure:

- i) To determine the resistivity at the site, electrical currents are induced into the ground through the use of electrodes.
- ii) soil resistivity is measured by finding the change in electrical potential between known horizontal distances within the electric field created by the current electrodes.
- iii) The four electrodes are placed in a straight line at equal distance ' D ' as shown in fig.



- iv) A direct voltage, causing current of 50 to 100 mA typically, is applied b/w the outer electrodes and potential drop is measured b/w the two linear electrodes.

v) Electrical resistivity is given by;

$$\rho = \frac{2\pi D E}{I}$$

Where; D = Distance b/n electrodes.

E = Potential drop b/n inner electrodes.

I = Current flowing b/n the outer electrodes

ρ = Mean resistivity.

Limitations:

1. This method is capable of detecting only the strata having different electrical resistivity.
2. The results may be influenced by surface irregularities, wetness of strata and salt concentration of ground water.
3. At the interface of strata resistivity do not change abruptly as assumed, so interpretation of results becomes difficult.
4. Skilled experts are needed at site.

3. Concept of sampling

12. Define disturbed sample and undisturbed sample. state the factors affecting on sample Disturbance.
13. Describe various types of soil samples. what is area ratio? state its significance.
14. Define area ratio, inside clearance, outside clearance.
15. Write short note on sampling tube.
16. Write short note on Stationary piston sampler.

* Types of soil samples

soil samples are used for determining engineering properties of the soils during sub-surface exploration. There are two types of soil samples.

i) Disturbed samples:

The samples in which the natural structure of the soil gets disturbed during sampling are known as 'disturbed samples'. However the mineral content of the soil and index properties of soil like grain size, plasticity, sp. gravity remain unchanged.

ii) Undisturbed samples:

The samples in which the natural structure of the soil and the water content are retained are known as undisturbed samples. Practically, it is almost impossible to get truly undisturbed sample. These samples are used for determining properties of soils such as shear strength, compressibility and permeability.

* Factors affecting the sample disturbance

- The disturbance of the soil samples depends up on the following design factors

1. Area ratio: The area ratio is defined as, The ratio of maximum cross sectional area of the cutting edge to area of the soil sample

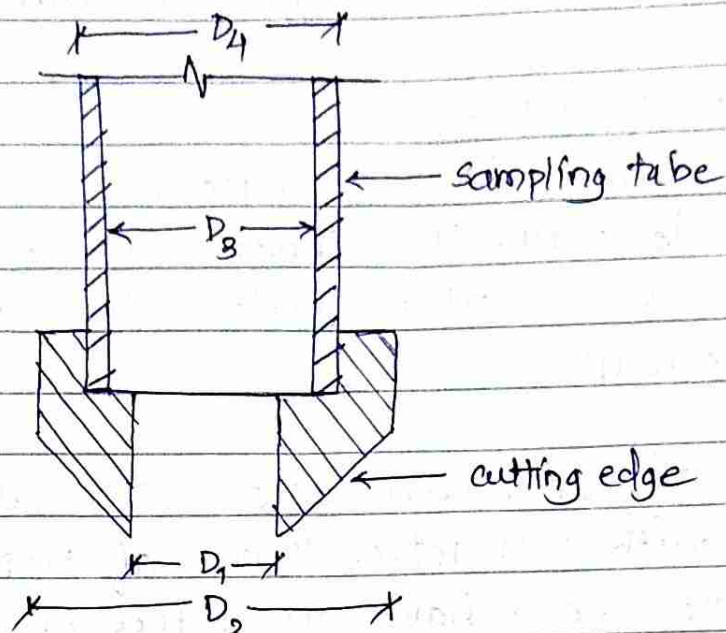
$$\text{i.e. } Ar = \frac{\text{Maximum c/s area of cutting edge}}{\text{Area of the soil sample}} \times 100$$

$$Ar = \left(\frac{D_2^2 - D_1^2}{D_1^2} \right) \times 100$$

Where;

$D_1 =$ Inner dia. of the cutting edge

$D_2 =$ outer dia. of the cutting edge.



Area ratio should be 10% or less, for obtaining good quality undisturbed samples.

2. Inside clearance:

It is defined as;

$$C_i = \frac{D_3 - D_1}{D_1} \times 100$$

Where; $D_3 =$ Inner dia. of the sampling tube.

Inside clearance should be 0.5 to 3% for obtaining good quality samples.

3. Outside clearance:

It is defined as;

$$C_o = \frac{D_2 - D_4}{D_4} \times 100$$

where; D_4 = outer dia. of the sampling tube.

The outside clearance should be as small as possible for reducing the driving force. Normally, outside clearance lies b/n 0 to 2%.

C_0 should not much greater than C_1 .

4. Inside wall friction:

Disturbance in the sample is caused by the friction on the inside wall. The inner surface of the sampler should be smooth, oil is applied to reduce friction on the inside wall.

5. Design of non-return valve: Generally, samplers are provided with non-return valve of proper design. Non-return valve should have an orifice of large area to allow air, water or slurry to escape quickly when the sampler is driven into the ground, while withdrawing sampler it should close immediately.

6. Method of applying force:

The degree of disturbance depends up on the following factors:

- The method of applying force during sampling &
- The rate of penetration of the sampler. For getting undisturbed samples, sampler should not be driven but should be pushed.

7. Recovery ratio: The recovery ratio is given by

$$R_r = \frac{L}{H}$$

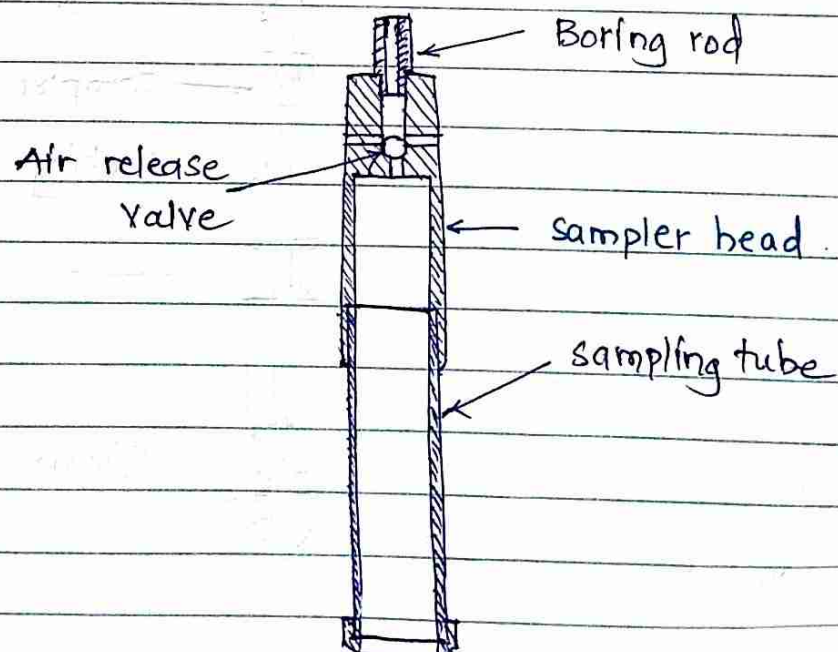
where; L = length of sample within the tube
 H = Depth of penetration of the sampling tube

The value of recovery ratio should be 96 to 98% for a satisfactory undisturbed sample.

* Sampling Tubes/samplers

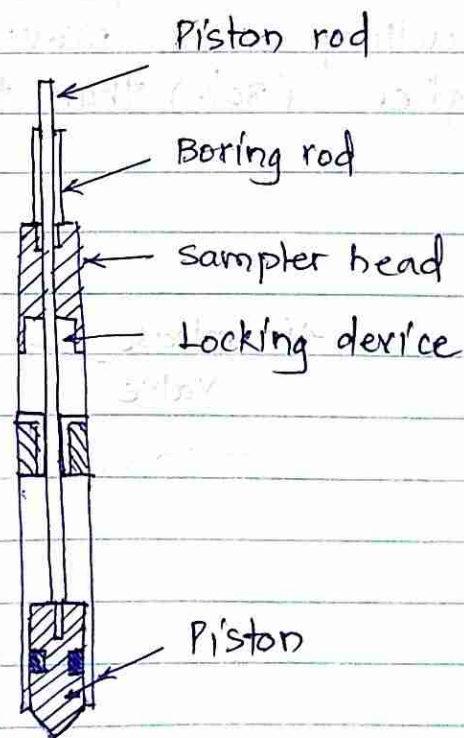
① Open Drive Sampler

- It consists of a steel tube with screw thread at each end. cutting shoe is attached to one end of the tube.
- The other end of tube screws into a sampler head to which the boring rods are connected.
- The sampler head also incorporates a non-return valve to allow air and water to escape as the soil fills the tube and to help retain the sample as the tube is withdrawn.
- The sampler is suited for clayey soil. when used to obtain samples of sand a short length of tube with spring loaded flaps should be fitted b/n the tube and cutting shoe to prevent loss of soil. Due to cutting shoe provided at bottom area ratio is higher (30%) than thin walled sampler.



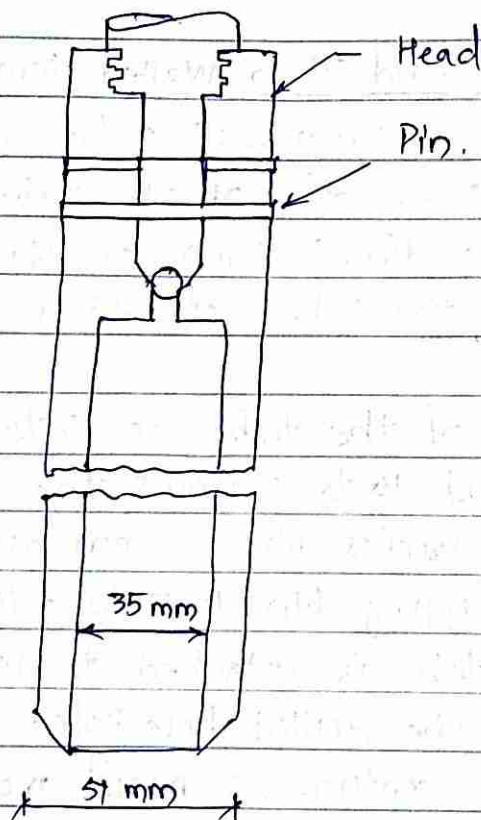
② Stationary Piston Sampler

- This sampler is first lowered into the bore hole, piston being located at the lower end of tube. The tube and piston could be locked together by a clamping device which is provided at the top of rods. Piston thus prevents water or loose soil from entering the tube.
- Now the piston is held against the soil by clamping the piston rod to the casing and the tube is pushed past the piston until the sampler head meets the top of the piston to obtain a sample.
- The sampler is then withdrawn holding the piston at the top of tube as this takes place. The vacuum b/w piston & sample held to retain the soil in the tube. Piston samplers are always pushed by mechanical or hydraulic jacking and they are not driven. This sampler is generally used for soft clays.



③ Split spoon sampler / spit barrel

- This sampler is used in standard penetration test tube which splits longitudinally into two halves. There is considerable disturbance in the sampler and useful in sandy type of soils. outside and inside dia. of tube is 51 mm and 35 mm. Area ratio is 112%.



④ Rotary sampler

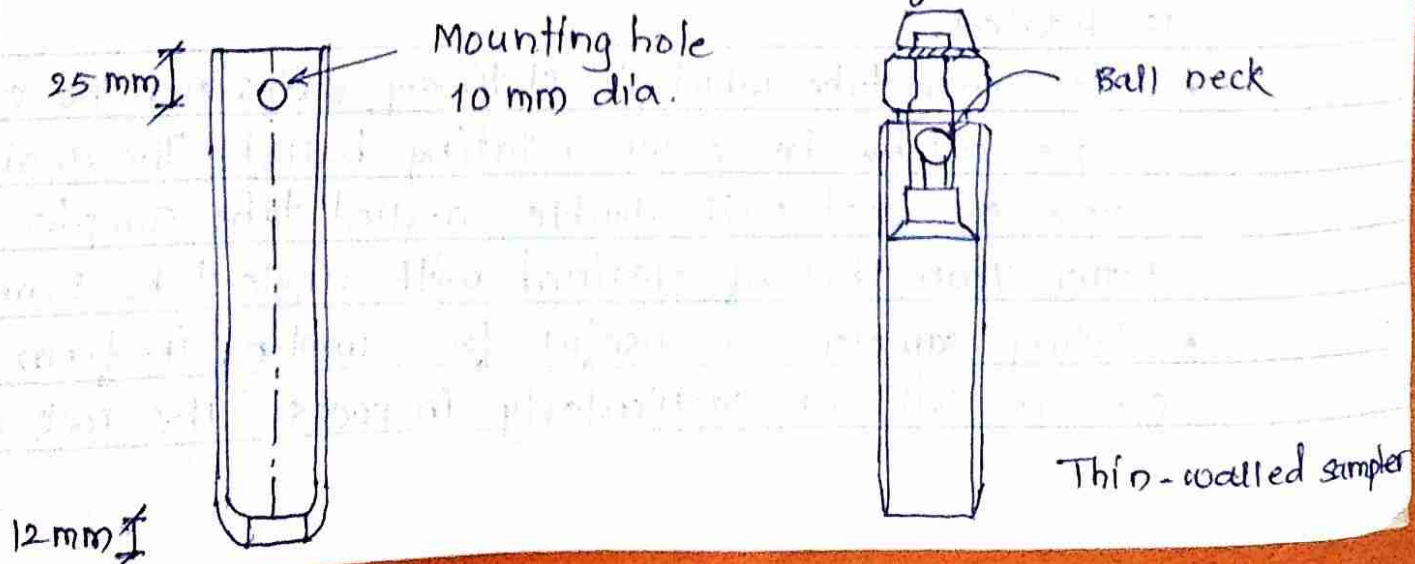
- Rotary sampler is double-walled tube sampler with an inner removable liner. The outer tube is provided with a cutting bit. The bit cuts an annular ring when the barrel is rotated.
- The inner tube which is stationary, slides over the cylindrical sample cut by the outer rotating barrel. The quality of sample obtained with double walled tube sampler is better than that of obtained with single tube sampler.
- Rotary samplers are useful for sampling in firm to hard cohesive soil and particularly in rocks. The rock quality

can be estimated from the core recovery ratio termed as rock quality designation RQD.

- The ratio of the total length of core recovered to the length of sampler advanced on a given run, expressed as percent, is the value of RQD. While determining the length of core recovered, only those piece of core which are atleast 100mm long, hard and sound, are considered.

⑤ Shelby Tubes and Thin Walled samplers:

- Shelby tubes are thin wall tube samplers made of seamless steel. The outside dia. of the tube is 40 mm to 125 mm. Samplers having dia. 50.8 mm or 76.2 mm are commonly used. The bottom of the tube acts as a cutting edge, which is sharpened.
- The length of the tube is 5 to 10 times the dia. for sandy soils and 10 to 15 times the dia. for clayey soils. The dia. generally varies b/n 40 mm and 125 mm and thickness of tube is ranging b/n 1.25 to 3.15 mm.
- The sampler tube is attached to drilling rod and lowered to the bottom of the drilled bore hole. Then, tube is pushed into the soil. By a continuous rapid motion without impact or twisting tube is pushed into the soil. The tube should be pushed to the length provided for the sample. After at least 5 min. the tube is turned twice through 360° to shear the sample off at the bottom before it is taken out.



4. Field Tests

Que: 17 Discuss SPT and what are the various corrections? What is the importance of the test in geotechnical engineering?

18. Discuss the pressure meter test with diagram.

19. Write short note on: DCPT, SCPT

20. Differentiate b/n

a) SPT and DCPT

b) BCT and DCPT

→ The following penetration tests are commonly used in practice -

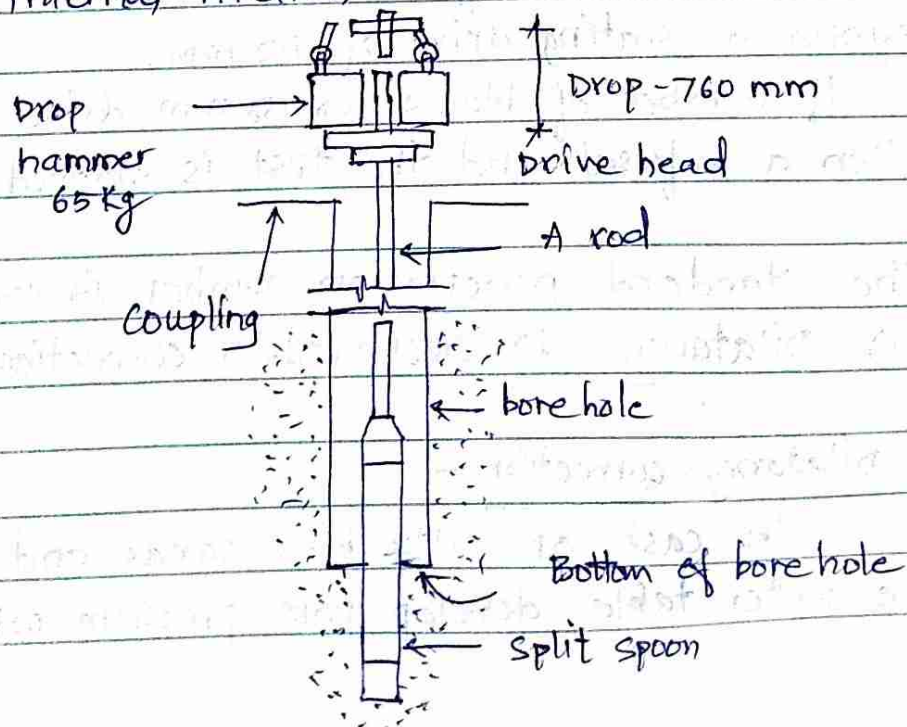
i) Standard Penetration Test (SPT)

ii) Dynamic Cone Penetration Test (DCPT)

iii) Static Cone Penetration Test (SCPT) or Dutch Cone Test (DCT)

i) Standard Penetration Test (SPT)

- SPT test is widely and commonly used to determine the parameters of the soil on site (In-situ)
- This test is especially useful for cohesionless soils as a correlation has been established between the SPT value & angle of internal friction of the soil.



SPT can be described as follows -

- (i) The split spoon sampler is used for conducting SPT test.
- (ii) After drilling a bore hole to the desired depth, the drilling tools are removed.
- (iii) After removing drilling tools, the sampler is lowered to the bottom of the hole.
- (iv) A hammer of 640 N (65 kg) weight with a free fall of 750 mm is used to drive the sampler.
- (v) The hammer is dropped at the rate of 30 blows/min.
- (vi) The number of hammer blows required to drive 150 mm of the sample is counted.
- (vii) The sampler is further driven by 150 mm and the number of blows is recorded.
- (viii) The sampler is once again further driven by 150 mm & number of blows counted.
- (ix) The number of blows recorded for first 150 mm is disregarded.
- (x) The number of blows required for last two 150 mm intervals are added to give standard penetration number (CN)

In other words, the std. penetration number (N) is equal to number of blows required for 300 mm of penetration beyond a seating drive of 150 mm.

If number of blows for 150 mm drive exceeds 50, it is taken a refusal and the test is stopped.

The standard penetration number is corrected for;

- i) Dilatancy
- ii) Overburden correction.

i) Dilatancy correction -

In case of silty fine sands and fine sands below the water table develop pore pressure which is not easily

dissipated. The pore pressure increases the resistance of the soil and hence the penetration number (N)

Following correction is recommended in case of silty sands when observed value of N' exceeds 15.

$$N = 15 + \frac{1}{2} (N' - 15)$$

where; N' = observed SPT value

N = corrected SPT value

where; $N' \leq 15$, $N = N'$

(ii) Overburden correction -

- In granular soils, the overburden pressure affects the penetration resistance. For SPT made at shallow depth, the values are too low. At a greater depth, same soil, at the same density index, would give higher penetration resistance.

Therefore, the effect of overburden pressure on SPT value may be approx. by Gibbs equation,

$$N = N' \times \frac{350}{(\sigma + 70)} \quad \left| \quad \begin{array}{l} \text{OR } N' = 0.77 \times \log_{10} \frac{2000}{\sigma} N_{ob} \\ N' = \text{corrected } N' \text{ values.} \end{array} \right.$$

where; σ = Effective overburden pressure in KN/m^2 not exceeding 280 KN/m^2

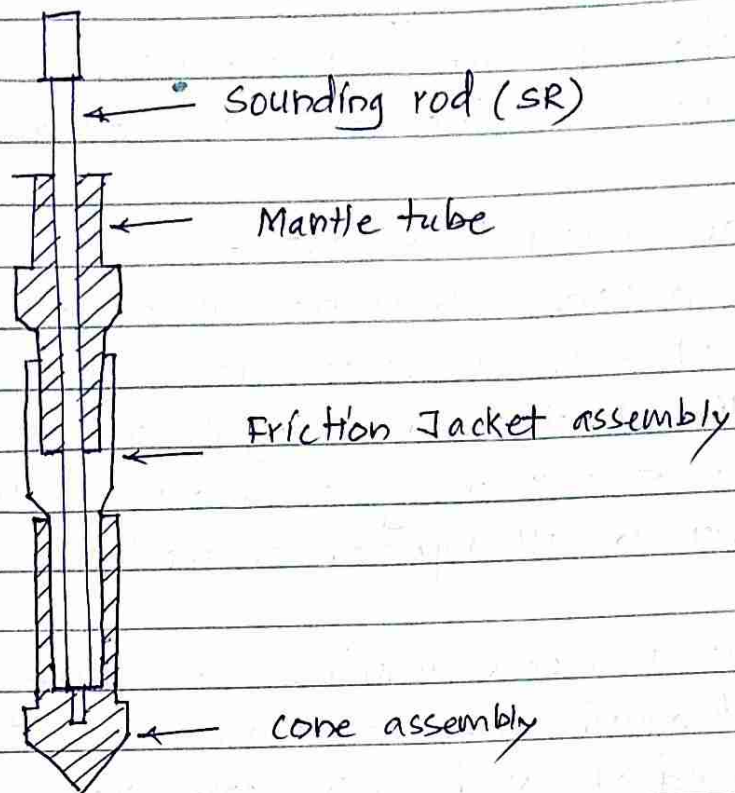
= depth $\times$ density

($\sigma \leq 280 \text{ KN/m}^2$ & $N \leq 2N'$)

Limitations of SPT:

- As SPT subjected to many errors SPT value is not very reliable.
- Corrections for overburden and dilatancy effect are to be applied to record ' N ' values.
- SPT value is not appropriate for determining the bearing pressure in fine-grained cohesive soil.
- Determination of allowable bearing pressure from ' N ' values is not accurate.

* Static Cone Penetration Test (SEPT)



1. Initially the cone and Friction jacket assembly is in a stationary position.
2. The cone is then pushed into the ground (soil) at the rate of 20mm/sec by the inner sounding rod at steady rate.
3. The force Q_c offered by soil for penetration is measured on pressure gauge. $q_c = Q_c/A_c$, where A_c is the base area.
4. The sounding rod is pushed further pushing the Friction jacket and the cone assembly together. The total force Q_t required for this is again read on the pressure gauge.

$Q_t = Q_c + Q_f$ where; Q_f = force required to push Friction jacket.

From this skin friction f_s is calculated.

$$f_s = \frac{Q_f}{A_f} \quad \text{where; } A_f \text{ is the surface area of Friction jacket.}$$

5. The cycle is repeated till desired depth is reached.
Thus, with this test, point resistance q_p as well as skin resistance Friction F_s can be found out.
6. This test is suitable for sandy or gravelly soil.

* Dynamic cone penetration test -

1. In this test, a cone which has an apex angle of 60° & attached to drill rods is driven into soil with a hammer of 65 kg, falling freely from a height of 750 mm.
2. The blow count for every 100 mm is continuously recorded. The cone is driven till refusal or up to the required depth.
3. The number of blows required for 300 mm penetration is noted as dynamic cone resistance. This test gives a continuous record of Ncd with depth. No sample however can be obtained in this test.
4. Dynamic cone penetration tests are performed either by using 50 mm dia. cone or 65 mm dia. cone.
5. The results obtained from DCPT can be compared with the SPT data.
- 6.

When 50 mm dia. cone is used

$N_{cd} = 1.5 N$ for depth up to 3m

$N_{cd} = 1.75 N$ for depth from 3m to 6m

$N_{cd} = 2.0 N$ for depths greater than 6m.

When 65 mm dia cone is used

$N_{cd} = 1.5 N$ for depth up to 4m

$N_{cd} = 1.75 N$ for depths from 4m to 9m.

$N_{cd} = 2.0 N$ for depths greater than 9m.

* Differentiate b/n SPT and DCPT

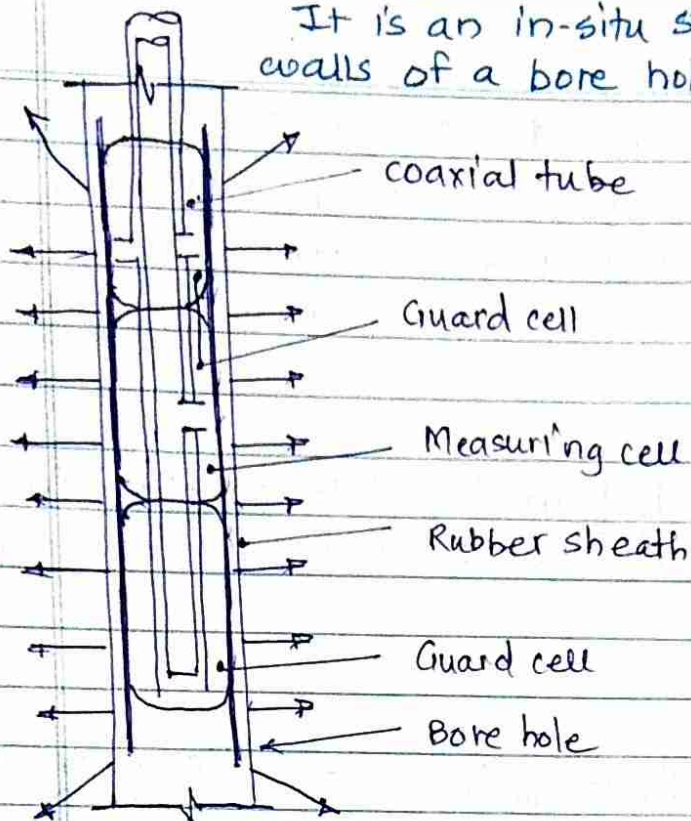
Sr.No.	SPT	DCPT
01.	This test is carried out in borehole.	01. DCPT can be carried out without a bore hole
02.	Split spoon sampler is attached to drill rod	02. No sample is taken. cone of apex angle 60° is attached to drill rod.
03.	SPT is costly and consume time more than DCPT	03. DCPT is less expensive & quicker than SPT
04.	Sample is extracted in SPT	04. No sample is extracted.
05.	Test not suitable in gravelly soils	05. DCPT can be carried out in gravelly soils also.
06.	corrections to recorded blow count required.	06. Dynamic cone resistance value is directly measured.

* Differentiate b/n SCPT and DCPT

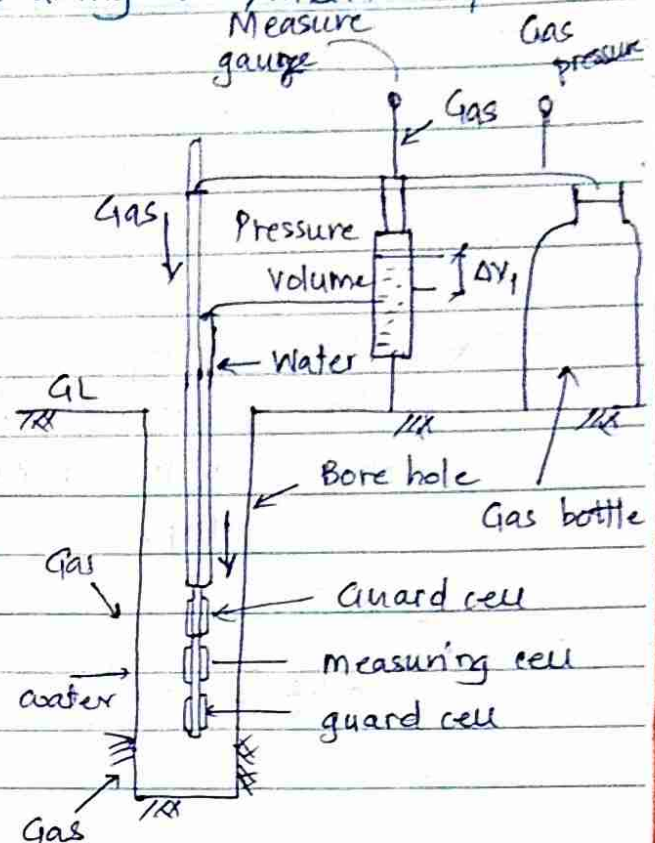
	SCPT	DCPT
01.	This test is suitable for soft clays and silts and fine to medium sand.	01. This test is suitable in dense sands and in hard and stiff clays.
02.	cone is pushed gradually into the ground at rate of 20 mm/sec and pressure on rod is measured at top	02. cone is driven by hammering and penetration resistance is measured in terms of number of blows. Ncd for 300 mm penetration.
03.	Measures tip resistance & skin friction.	03. Measures only tip resistance
04.	setup is very simple & portable	04. setup is complicated require more efforts.
05.	Test is rapid & less costly	05. more costly than SCPT.

4) Pressuremeter Test -

It is an in-situ stress-strain test performed on the walls of a bore hole using a cylindrical probe.



Pressure cell



The test results are used either directly or indirectly for the design of foundations.

Procedure -

- i) The probe is inserted into the bore hole.
- ii) The probe is inflated by applying water pressure and gas.
- iii) The probe presses against the unlined walls of the bore hole.
- iv) The pressure in guard cell is kept equal to pressure in probe or measuring cell.
- v) As the pressure increases soil deforms.
- vi) The volumetric deformation of the bore hole is obtained by noting pressure in the probe.
- vii) The pressure deformation data obtained from the test may be used for determining modulus of deformation, untrained shear strength angle of shearing resistance and other engg. properties of soil.

30

5. Concept of core logging and site investigation report.

Que 21 Explain with sketch a typical "Core Log". Indicate core recovery and RQD values for various rock types.

22. write short note on presenting report on site investigation

→ * core logging -

- * core logging -
 - The results obtained from base hole are plotted in a form, with which nature of soil profile can be made it's called as core logging.

• Example:

[illegible]

- such a presentation gives information like type of soil met at various depth, such as sand, silt, clay etc. Depth at which ground water table is met, values of SPT test or vane shear test nature of sample collected, i.e. whether disturbed or undisturbed nature of rock sample viz. Fractional / weathered RQD etc.

- Results of various bore holes taken are plotted on single sheet with suitable scale.

- When it's not possible for such layers, it's an indication that strata is erratic or there are pockets of thin layers. Such information is very useful in deciding depth of formation and bearing capacity to be assumed.

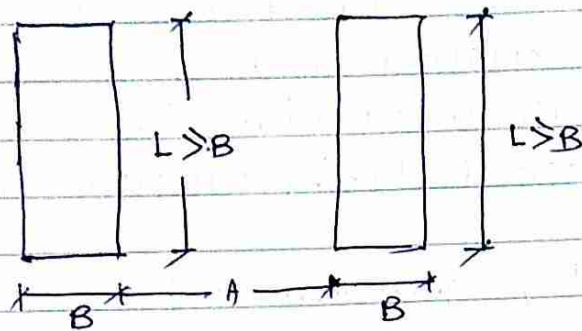
* Site investigation report

- The report should include, aim of project and what the client expects to explore.
- Methodology - Methodology adopted to achieve the goals aimed at should be explained in brief.
- Presenting results of site investigation: This should include presenting bore hole data including soil profile, properties of soil.
- Analysis of site tests, conclusions and possible solutions/recommendations.

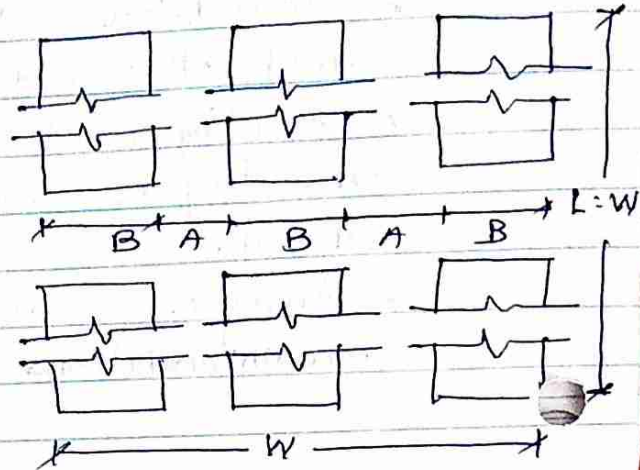
Que-23 Discuss how you will determine the depth and number of boreholes for important projects.

- Exploration, in general, should be carried out to a depth up to which the increase in pressure due to structural loading is likely to cause perceptible settlement or shear failure.
- Such a depth, known as the significant depth, depends up on the type of structure, its weight, size, shape and disposition of the loaded areas, and the soil profile & its properties.
- It is generally safe to assume the significant depth up to a level at which the net increase in vertical pressure becomes less than 10% of the initial overburden pressure.
- Alternatively, a pressure bulb bounded by an isobar of one-fifth or one-tenth of the surface loading intensity is sometimes assumed to define the minimum depth of exploration.
- This depth may be assumed to be equal to one-and-a-half to two times the width of loaded area.

- The depth of exploration at the start of the work may be decided according to the following guideline given in IS 1892 - 1979.



Isolated spread footings / Rafts



Adjacent rows of footings.

A = clear spacing b/w footing

B = width of footing

L = length of footing $> B$

Sr.No.	Type of foundation	Depth of exploration (D)
01.	a) Isolated spread footing or raft	clear span (A) D
		$A > 4B$ $1.5B$
	b) Adjacent footing	$A < 2B$ $1.5L$
02.	Adjacent rows of footing	$A < 2B$ $4.5B$
		$A > 2B$
		$< 4B$ $1.5B$
		$A \geq 4B$
03.	Pile & well foundation	i) 1.5 width of structure ii) up to depth of scour iii) $10m$.
04.	Dams	$1.5 \times$ bottom width of dam
05.	Black cotton soil	$3.5m$

Que 24 Describe extent of exploration

- • Number & disposition of pits and bore holes should be such as to reveal any major changes in thickness, depth or properties of strata and immediate surroundings.
- If ground conditions are uniform, min. 3 holes not in a line are enough to give a reliable soil profile.
 - For building project, 4 holes at corners and one at centre of the area are recommended.
 - For large area, one hole per 250 sq.m or holes spaced at 15-20m may be adequate or other sounding tests at every 100m.
 - For smaller, less important buildings one bore hole or test pit in centre may suffice.
 - For dam sites, preliminary borings at 50m spacing along the top line of upstream face of dam and across one or both abutments. Few widely scattered holes are also required in bottom of reservoir on upstream side of dam.
 - In case of road sites, pits or bore holes along proposed centre line of road along ditch lines alternating right and left ditches. Spacing at 100m at start and may increase to 500m for uniform soil or decreased to 30m or less for erratic deposits.
 - In erratic deposits, increase in number of holes does not necessarily improve quality of information. In such cases less expensive method of exploration such as sounding should be employed.

Numerical Examples

Ex. 1

Determine area ratios of samplers of the following description.

- i) Split spoon sampler, $D_o = 50 \text{ mm}$ and $D_i = 36 \text{ mm}$
- ii) Drive tube, $D_o = 100 \text{ mm}$ and $D_i = 85 \text{ mm}$.
- iii) Shelby tube, $D_o = 50 \text{ mm}$ and $D_i = 45 \text{ mm}$.

→ Given;

As above;

We know that;

$$Ar = \frac{D_o^2 - D_i^2}{D_i^2} \times 100 \quad \text{OR} \quad \frac{D_o^2 - D_i^2}{D_i^2} \times 100$$

 D_o or D_2 = outer diameter D_i or D_1 = Inner dia.

- i) Split-spoon sampler

$$Ar = \frac{(50)^2 - (36)^2}{(36)^2} \times 100 = 92.90\% \text{ (More disturbed)}$$

- ii) Drive tube

$$Ar = \frac{(100)^2 - (85)^2}{(85)^2} \times 100 = 38.40\% \text{ (Fairly disturbed)}$$

- iii) Shelby tube

$$Ar = \frac{(50)^2 - (45)^2}{(45)^2} \times 100 = 23.45\% \text{ (Undisturbed sample)}$$

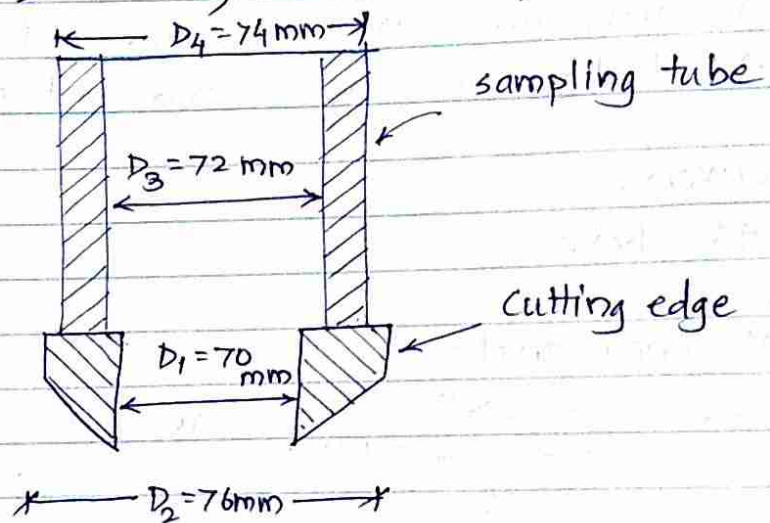
Ex-2

The inner dia. of sampling tube and that of cutting edge are 72 mm and 70 mm respect. Their outer dia. are 74 mm and 76 mm respect. Determine the inside

clearance, outside clearance and area ratio of sampler.

→ Given:

$$D_1 = 70 \text{ mm}, D_2 = 76 \text{ mm}, D_3 = 72 \text{ mm}, D_4 = 74 \text{ mm}$$



$$\text{Area ratio, } A_r = \frac{D_2^2 - D_1^2}{D_1^2} \times 100$$

$$= \frac{(76)^2 - (70)^2}{(70)^2} \times 100 = \underline{\underline{17.87\%}}$$

$$\text{outside clearance} = \frac{76 - 74}{74} \times 100 = 2.70\%$$

$$\text{Inside clearance} = \frac{72 - 70}{70} \times 100 = 2.86\%$$

Ex-3 Find the corrected blow count at 12m depth of unit wt. of soil = 18 kN/m^3 . Blow count observed in SPT test is 25. Water table is at a depth of 2m and unit wt. of soil above GWT is 16 kN/m^3 .

→

$$\begin{aligned} \text{Total stress, } \sigma \text{ at 20m depth} &= (18 \times 12) + (16 \times 2) \\ &= 212 \text{ kN/m}^2 \end{aligned}$$

$$\text{Pore pressure, } u \text{ at 20m depth} = 10 \times 10 = 100 \text{ kN/m}^2$$

Effective stress, σ' at 20m depth = 112 kN/m²

$\therefore$ Effective stress = (Total stress - Pore water pressure)

$$N' = \left[0.77 \log \frac{2000}{112} \right] \times 25$$

$$= 24$$

correction for dilatancy $\cdot N'' = 15 + \left(\frac{24-15}{2} \right) = 19.5 \approx \underline{20}$

Ex: 4 Calculate the depth of soft strata underlain by hard strata. Use the following data:

$V_1 = 600$ m/sec and $V_2 = 3000$ m/sec and the break in the plot was located at 60m.

→

$$V_1 = 600 \text{ m/sec} \quad V_2 = 3000 \text{ m/sec} \quad d_c = 60 \text{ m}$$

Depth of soil strata; $H_1 = \frac{d_c}{2} \sqrt{\frac{V_2 - V_1}{V_2 + V_1}}$

$$H_1 = \frac{60}{2} \sqrt{\frac{3000 - 600}{3000 + 600}} = \underline{24.5 \text{ m}}$$

Ex: 5 In a geophysical exploration, the time distance plot gave $V_1 = 300$ m/sec, and $V_2 = 900$ m/sec and the break in the plot was located at 35m. Determine the depth of over burden.

→

$$V_1 = 300 \text{ m/sec} \quad V_2 = 900 \text{ m/sec} \quad d_c = 35 \text{ m}$$

$$H_1 = \frac{d_c}{2} \sqrt{\frac{V_2 - V_1}{V_2 + V_1}}$$

$$= \frac{35}{2} \sqrt{\frac{900 - 300}{900 + 300}}$$

$\therefore$ Depth of over burden = 12.37 m,

Ex: 06 calculate corrected blood count for recorded blood count of 32 and comment on results.

→ $N' = 32$

$$N'' = 15 + \left(\frac{N' - 15}{2} \right) = 15 + \left(\frac{32 - 15}{2} \right) = 23.5$$

Ex: 07 A sampling tube of 15 cm internal diameter is 1 m thick. It is fitted with a cutting edge. The inside dia. of cutting edge is flushed with sampling tube. The cutting edge is 1.2 mm thick. Compute inside clearance, outside clearance and area ratio. comment on sample collected by this tube.

→

cutting edge, Internal dia., $D_1 = 150 \text{ mm}$

Ext. dia., $D_2 = 152.4 \text{ mm}$

$$\begin{aligned} \text{Area ratio} &= \frac{D_2^2 - D_1^2}{D_1^2} \times 100 \\ &= 3.22\% < 10\% \end{aligned}$$

Sample is in undisturbed state

sampling tube, Internal dia. $D_3 = 150 \text{ mm}$

$$\begin{aligned} \text{Ext. dia.} \quad D_4 &= 150 + 1 + 1 \\ &= 152 \text{ mm} \end{aligned}$$

$$\text{Inside clearance} = \frac{D_3 - D_1}{D_1} \times 100 = 0$$

$$\text{outside clearance} = \frac{D_2 - D_4}{D_4} \times 100 = 0.26\%$$

Sample collected is undisturbed sample.